

T01 THERMAL PERFORMANCE | P

Intent : Provide a thermal environment that the majority of building users find acceptable.

Summary : This WELL feature requires projects to create indoor thermal environments that provide comfortable thermal conditions to the majority of people in support of their health, well-being and productivity.^{1,2}

Issue : The thermal environment substantially impacts a building's energy footprint, since in many countries heating and cooling account for approximately half of a building's energy consumption.^{3,4} Furthermore, the indoor thermal environment is ranked as one of the strongest contributing factors to overall human satisfaction in the built environment.⁵ The thermal environment not only impacts comfort and productivity, but due to its linkages to integumentary, endocrine and respiratory body systems, thermal comfort can also cause a variety of detrimental health outcomes.^{1,6} Overly warm indoor spaces are linked to increases in sick building syndrome symptoms, irregular heart rate, respiratory issues, fatigue and negative mood.⁷ Cold work environments have been linked to increased effort and work towards maintaining proper posture and increased risk for chronic issues related to musculoskeletal health.⁸

Solutions : Thermal comfort standards utilize a model that provides a means of predicting whether humans in a mechanically conditioned space will be satisfied with the thermal environment based on six core parameters: air temperature, humidity, air movement, mean radiant temperature of surrounding surfaces, metabolic rate and clothing insulation.^{9,10} For naturally conditioned buildings, the adaptive thermal comfort model correlates human comfort directly with indoor operative temperature and outdoor temperature.^{11,12} Achieving thermal satisfaction among people requires some level of control over thermal comfort parameters in any given environment.

PART 1 PROVIDE ACCEPTABLE THERMAL ENVIRONMENT

For All Spaces except Commercial Kitchen Spaces:

Option 1: Performance verified environmental conditions

The following requirements are met, as applicable:

- Mechanically conditioned regularly occupied spaces maintain thermal comfort conditions of PMV +/- 0.5 for at least 90% of occupied hours in at least 90% of regularly occupied spaces.¹³
- Naturally conditioned regularly occupied spaces meet all the following conditions.¹²

	Prevailing Mean Outdoor Temperature, $t_{pma(out)}$	Indoor Operative Temperature	Notes
Minimum	10 °C	$t_{pma(out)} \times 0.31 + 14.3$ °C	N/A
Maximum	33.5 °C	$t_{pma(out)} \times 0.31 + 21.3$ °C	Occupant-controlled elevated air speed may be used to increase this maximum per ASHRAE 55

- Mixed-mode-conditioned spaces meet the requirements for both mechanically and naturally conditioned spaces, when each is in operation

Note : Spaces with multiple occupant types may use the occupant characteristics of PMV—metabolic rate and clothing value—for the majority occupant type (e.g., a space with 30 diners and 5 servers may use the diners' metabolic rates and clothing values to determine the PMV for that space).

OR

Option 2: Long-term thermal data

The following requirements are met:

- Project meets Feature T06 Thermal Comfort Monitoring.
- Sensor data demonstrate parameters meet one of the following:
 - One of the PMV or temperature ranges described in Option 1. Dry bulb temperature may be used in place of operative temperature. Naturally conditioned projects must also measure outdoor air temperature.
 - Dry bulb temperature is between 21-25 °C for occupied hours.¹⁴ The designed air velocity is not more than 0.2 m/s at 1.7 m above the floor.

Note : Spaces with multiple occupant types may use the occupant characteristics of PMV—metabolic rate and clothing value—for the majority occupant type (e.g., a space with 30 diners and 5 servers may use the diners' metabolic rates and clothing values to determine the PMV for that space). Projects undergoing recertification which were previously awarded Feature T06 must consider all data collected since the previous certification.

OR

Option 3: Thermal comfort surveys

The following requirement is met:

- The project achieves at least two points in Feature T02: Verified Thermal Comfort.

For Commercial Kitchen Spaces:

The following requirement is met:

- The operative temperature in the kitchen does not exceed 27 °C.

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily

Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEASURE THERMAL PARAMETERS

For All Spaces except Dwelling Units:

Option 1: Semi-annual testing

The following requirements are met:

- a. The dry-bulb temperature, relative humidity, air speed (only for projects that use elevated air speed method) and mean radiant temperature are measured in regularly occupied spaces at intervals no longer than twice per year (including once in June, July or August and once in December, January or February), and the results are submitted annually through the WELL digital platform.
- b. The number and location of sampling points comply with the requirements outlined in the Performance Verification Guidebook.

Note: Projects are not required to follow the device requirements or test methods described in the Performance Verification Guidebook. Permanently installed monitors may be located on interior walls rather than a minimum distance away.

OR

Option 2: Continuous monitoring

The following requirement is met:

- a. Project meets Feature T06 Thermal Comfort Monitoring.

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T02 VERIFIED THERMAL COMFORT | O (MAX : 3 PT)

Intent : Enhance thermal comfort and promote human productivity, by ensuring that a substantial majority of building users (above 80%) perceive their environment as thermally acceptable

Summary : This WELL feature requires projects to provide high levels of thermal comfort, by determining occupant satisfaction through a survey.

Issue : Due to the strong influence of the thermal comfort, standards establish that building's thermal environment must satisfy a minimum of 80% of occupants to be considered acceptable.¹ However, despite having established standards, studies have also shown that only 11% of office buildings surveyed in the U.S. provide thermal environments that meet generally accepted goals of human satisfaction². Similarly, as many as 41% of office workers have expressed dissatisfaction with the thermal environment.^{2,3} Building users who are satisfied with their thermal environment have been shown to be more productive in the workplace,² while thermal discomfort is associated with sick building syndrome symptoms and other conditions that lead to a decrease in productivity.^{2,4}

Solutions : Building occupants are an invaluable source of information that can be used for improving the performance of buildings. Thermal comfort surveys allow projects to objectively gauge which building services and design features are or are not performing well and help to prioritize the steps needed to improve occupant thermal comfort satisfaction and workplace productivity. If survey results indicate that percentage of occupants dissatisfied with thermal conditions in the building exceeds the targeted thresholds, it is necessary to develop a detailed plan for action and commitment to address occupant dissatisfaction with thermal conditions.

PART 1 SURVEY FOR THERMAL COMFORT (MAX : 3 PT)

For All Spaces:

A post-occupancy survey is administered at least twice a year, including once in June, July or August and once in December, January or February. For buildings located in tropical regions, the survey may be administered twice a year, at least four months apart. Survey is administered at least six months after occupancy, which satisfies the following conditions:

- a. All regular occupants are invited to participate in the anonymous survey, and responses are collected from the following number of respondents:¹

Number of Regular Occupants	Minimum Number of Responses
More than 45	35% of those regular occupants
20 to 45	15 regular occupants
Less than 20	80% of those regular occupants

- b. The survey includes an assessment of overall satisfaction with thermal performance and identification of thermal comfort-related issues in accordance with either:
 1. The sample survey in Appendix T1.
 2. Any pre-approved survey listed in Part 1: Administer Project Survey in Feature C04: Occupant Survey.
- c. The results of the survey responses comply with one of the target satisfaction thresholds as specified in the table below:

Tier	Thermal Comfort Satisfaction Thresholds	Points
1	80% of regular occupants	2
2	90% of regular occupants	3

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T03 THERMAL ZONING | O (MAX : 2 PT)

Intent : Enhance thermal control of building occupants through the provision of thermal zones.

Summary : This WELL feature requires projects to increase thermal control of the space, by allowing control of either the conditions of a thermal zone or movement between thermal zones.

Issue : People in buildings who are satisfied with their thermal environment have been shown to be more productive in the workplace.^{1,2} For instance, employees perform on average 15% poorer in hot conditions and on average 14% poorer in cold conditions.³ An average 2% decrease in work performance per 1 °C of air temperature rise has been found, when the air temperature is above 25 °C.⁴ On the other hand, even a 1 °C decrease in air temperature within the range of 20 °C-25 °C is linked to increased student performance in mathematics.⁵ Furthermore, thermal comfort preferences are highly individual and can be affected by metabolism, body type, clothing and other personal factors.⁶⁻⁸ These personalized factors make it nearly impossible to find a temperature that will satisfy all individuals in the same space at the same time.⁹

Solutions : Indoor air temperature can be influenced by a number of factors, including the building physics and orientation, building location and structure, occupant density, ventilation strategy and mode of operation. In the majority of buildings, the HVAC system is responsible for controlling indoor air temperature and humidity. Where temperature zoning is an option, individually accessible thermostats that enable people to set their own thermal conditions independently of other zones should be used.

PART 1 PROVIDE THERMOSTAT CONTROL (MAX : 2 PT)

For All Spaces:

The following requirements are met for at least 90% of regularly occupied spaces:

- a. Control over temperature in the space is available through either:
 1. Thermostats present within the thermal zone.
 2. A digital interface accessible to occupants on a computer or phone.
- b. The maximum size of each thermal zone is 60 m² or 10 occupants, whichever is larger. Individual regularly occupied rooms are considered separate zones.

c.

Projects earn points based on the number of thermal zones:

Tier	Number of Zones		Number of Zones	Points
1	One per 60 m ²	OR	One per 10 occupants	1
2	One per 30 m ²	OR	One per 5 occupants	2

- d. Temperature sensors are positioned at least 1 m away from exterior walls, windows and doors, direct sunlight, air supply diffusers, mechanical fans, heaters or any other significant sources of heat or cold.

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T04 INDIVIDUAL THERMAL CONTROL | O (MAX : 3 PT)

Intent : Maximize and personalize thermal comfort among all individuals.

Summary : This WELL feature requires projects to improve thermal comfort of people in the space through the provision of personal thermal comfort devices and flexible dress codes that support individual thermal preferences.

Issue : Many thermal comfort standards (such as ASHRAE 55) aim to provide a thermal environment, where at minimum 80% of people will perceive the thermal environment as acceptable.¹ Due to the individualized nature of thermal comfort, not everyone will be equally comfortable under the same environmental conditions. Besides the six primary factors of thermal comfort, many secondary factors, such as age, sex, health condition, personal thermal adaptation and thermal history, including climatological origin, are considered as major secondary factors.²⁻⁵ Factors, such as temperament, preferences, social and cultural norms and seasonal variation also play an important role in determining individual thermal comfort.^{6,7}

Solutions : Adopting holistic approaches through the provision of individual thermal control devices results in improved individual thermal comfort and in the expansion of the thermal comfort acceptability limits.^{6,8} Individual thermal control also allows for a broader range of recommended indoor air temperatures, which is linked to energy savings potential.⁹⁻¹¹ In larger shared spaces, use of personalized equipment is a good strategy to give people the ability to better control their sensation and comfort.¹²⁻¹⁴ For example, the use of a chair equipped with fans allows the room air temperature to increase without compromising thermal comfort.¹⁵⁻¹⁷ In addition, flexible dress codes are important, because clothing insulation is also one of the primary forms of intervention for addressing suboptimal thermal conditions.¹⁸

PART 1 PROVIDE PERSONAL COOLING OPTIONS (MAX : 1 PT)

For All Spaces except Dwelling Units:

The project provides all regular occupants with the ability to cool their individual environment through at least one of the following:¹

- a. A user-adjustable thermostat, which controls the environment for no more than one person and is connected to the building's mechanical cooling system or a more localized air conditioning unit.
- b. A desk, standing, pedestal or ceiling fan that does not increase air speed for other occupants.
- c. Chair with mechanical cooling system.
- d. Any other solution capable of affecting a PMV change of -0.5 within 15 minutes from activation, without changing the PMV for other occupants.

PART 2 PROVIDE PERSONAL HEATING OPTIONS (MAX : 1 PT)

For All Spaces except Commercial Kitchen Spaces & Dwelling Units:

The project provides all regular occupants with the ability to warm their individual environment through at least one of the following:¹

- a. A user-adjustable thermostat, which controls the environment for no more than one person and is connected to the building's mechanical heating system.
- b. Electric parabolic space heater.
- c. Electric heated chair or footwarmers.
- d. Personal or shared blankets. Shared blankets are washed or disinfected at least weekly after use.
- e. Any other solution capable of affecting a PMV change of +0.5 within 15 minutes from activation, without changing PMV for other occupants.

PART 3 ALLOW FLEXIBLE DRESS CODE (MAX : 1 PT)

For All Spaces except Commercial Kitchen Spaces & Dwelling Units:

The following requirement is met:

- a. A flexible dress code policy is in place that allows regular occupants to dress for individual thermal preferences.¹⁹

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T05 RADIANT THERMAL COMFORT | O (MAX : 2 PT)

Intent : Maximize volume of the space, reduce dust transmission, improve ventilation control and increase thermal comfort by incorporating radiant heat and cooling systems into the building design.

Summary : This WELL feature requires projects to use radiant systems and independently controlled ventilation systems.

Issue : Mean radiant temperature is one of the six core thermal comfort parameters. It is influenced by a surface material's ability to absorb or emit radiant heat, the extent to which the surface area is exposed to the person (view factor) and the temperatures of the surrounding objects. Uniform thermal radiation may be disrupted by cold windows, uninsulated walls, equipment and improperly sized heating panels, causing local thermal discomfort.¹ In addition, conventional "all-air" systems have a higher risk of draft discomfort, due to elevated indoor air velocity.² Studies show occupants have a 50% probability of indoor temperature satisfaction in spaces with radiant systems, compared to conventional "all-air" systems.³

Solutions : Radiant heating systems are designed to affect mean radiant temperature, and thus the heat exchange with the people in the space, by supplying heat directly to the surrounding surfaces of the floors, walls and ceilings. With radiant systems, the heat transfer due to radiation increases, while the heat transfer due to convection decreases. Buildings with radiant systems have the advantage of quiet operation, low energy consumption and the capability of design integration with an independent ventilation system.⁴ Coupling a radiant system with a dedicated air system separates the twofold role of the mechanical system controlling both heating/cooling and ventilation allowing for better control of the thermal environment. The use of radiant heating and cooling also reduces the number of allergens circulated in the air, since this type of system does not use forced air to distribute heating or cooling.

PART 1 IMPLEMENT RADIANT HEATING (MAX : 1 PT)

For All Spaces:

At least 50% of the regularly occupied project area is heated with one or more of the following:

- a. Radiant ceilings, walls or floors.
- b. Radiant panels which cover at least half of the wall or ceiling to which they are attached (does not include steam radiators).

PART 2 IMPLEMENT RADIANT COOLING (MAX : 1 PT)

For All Spaces:

At least 50% of the regularly occupied project area is cooled with one or more of the following:

- a. Radiant ceilings, walls or floors.
- b. Radiant panels that cover at least half of the wall or ceiling to which they are attached.

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T06 THERMAL COMFORT MONITORING | O (MAX : 1 PT)

Intent : Monitor and effectively address unacceptable thermal comfort conditions and inform building managers and users of the thermal comfort parameters of their indoor environment.

Summary : This WELL feature requires projects to monitor thermal comfort parameters using sensors in their buildings that can be used as feedback for building managers and users to take appropriate actions.

Issue : Unfavorable levels of heat, humidity and ventilation are associated with people's experience of itchy eyes, headache and throat irritation.¹ Outdoor weather, indoor occupancy and building physics and performance, including ventilation rates, are highly variable and have a direct impact on human perceptions of thermal comfort. To maintain ideal performance metrics, projects should continuously gather data on thermal comfort parameters in order to inform remediation actions.

Solutions : Building HVAC systems should be designed to monitor and control for variations in indoor air temperature, mean radiant temperature, relative humidity and air movement. Thermal comfort monitoring can help building users to be aware of and promptly fix any deviations in thermal comfort metrics. These measures by themselves will not resolve the issue of potential thermal discomfort, but they certainly raise awareness and are an important first step toward a solution. In addition to having calibrated sensors, the positioning of the sensors plays an important role in the accurate assessment of the thermal environment.

PART 1 MONITOR THERMAL ENVIRONMENT (MAX : 1 PT)

For All Spaces:

1: Thermal comfort monitors

The following requirements are met:

- a. The project monitors dry-bulb temperature and relative humidity in occupiable spaces in compliance with the requirements outlined in the Continuous Monitoring Protocols of the Performance Verification Guidebook.
- b. The thermal comfort data monitored is made available to occupants through one of the following:
 1. Display screens, with at least one screen located in each 500 m² zone of regularly occupied space.
 2. A website or mobile application, with at least one sign located in each 500 m² zone of regularly occupied space, indicating where the data may be accessed.
- c. The thermal comfort data are updated at least once every 15 minutes.

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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T07 HUMIDITY CONTROL | O (MAX : 1 PT)

Intent : Limit the growth of pathogens, reduce off-gassing and maintain thermal comfort by providing the appropriate level of humidity.

Summary : This WELL feature requires projects to maintain optimum relative humidity levels that are conducive to human health and well-being.

Issue : When air temperature is within a comfortable range for occupants, the effects of humidity on thermal comfort are generally unimportant.¹ However, in warm temperature settings, humidity can influence the degradation of building materials, the ability of the human body to release heat through evaporation and the level of discomfort from excess moisture on the skin.²⁻³ If the humidity is too high, the human body has a limited capacity to cool down through sweating.⁴ Warm and humid indoor spaces are also associated with mold and fungal growth.⁵ Moreover, humidity in warm spaces may promote the accumulation and growth of microbial pathogens, including bacteria and dust mites which can lead to odors and cause respiratory irritation and allergies in sensitive individuals.⁵ Conversely, cold and dry spaces can lead to discomfort and irritation of the airways, skin, eyes, throat and mucous membranes⁶ and facilitate the spread of the influenza virus.^{7,8}

Solutions : Buildings situated in climates with broad humidity ranges can maintain relative humidity within healthy and comfortable levels, by adding or removing moisture from the air.⁹

PART 1 MANAGE RELATIVE HUMIDITY (MAX : 1 PT)

For All Spaces:

Option 1: Mechanical humidity control

The following requirement is met in all regularly occupied areas, except high-humidity areas:

- a. The mechanical system has the capability of maintaining relative humidity between 30% and 60% at all times by adding or removing moisture from the air.^{10,11}

OR

Option 2: Humidity modeling

The following requirement is met for all regularly occupied areas, except high-humidity spaces:

- a. The modeled relative humidity levels in the space are between 30% and 60% for at least 98% of all business hours of the year.

OR

Option 3: Long-term humidity data

The following requirements are met:

- a. Project meets Feature T06: Thermal Comfort Monitoring.
- b.

Relative humidity levels in regularly occupied areas, except high-humidity spaces, are between 30% and 60% during occupied hours.

Note : Projects undergoing recertification, which were previously awarded Feature T06, must consider all data collected since the previous (re)certification

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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T08 B ENHANCED OPERABLE WINDOWS | O (MAX : 1 PT)

Intent : Provide the benefits of increased outdoor air supply while minimizing any resulting thermal discomfort.

Summary : This WELL feature requires windows that can be opened at different elevations to provide desired air flow at different outdoor temperatures.

Issue : Although use of personally controlled operable windows can improve satisfaction with the indoor air quality and thermal comfort, it may also have negative aspects if outdoor air is polluted or if outdoor weather is very hot or cold.¹⁻³ In cold weather, large openings in windows near occupants can lead to unpleasant draft.

Solutions : Windows which can be opened in multiple modes may provide thermal comfort over a variety of weather conditions. Large openings near the level of the occupants can provide high levels of air exchange and potentially increased air speed and corresponding cooling effect. Small openings near the ceiling allow some air exchange to take place while minimizing the potential for draft, allowing the outdoor air to warm before it reaches the occupants.⁴

PART 1 PROVIDE WINDOWS WITH MULTIPLE OPENING MODES (MAX : 1 PT)

Note : Projects may only achieve this part if Feature A07 Part 1 is also achieved.

For All Spaces:

1: Window design

Operable windows may be opened according to the following requirements (windows which may be opened in both modes may count for both requirements a and b):

- a. At least 70% of operable windows may be opened such that at least half of the opening is not more than 1.8 m above the finished floor and opening is at least 0.3 m in the smallest dimension. At least one such window is present in each room with operable windows.
- b. If project is equipped with heating, at least 30% of operable windows may be opened such that entirety of opening is at least 1.8 m above the finished floor (preferably as close to the ceiling as possible).⁵ At least one such window is present in each room with operable windows.
- c. Controls for window operation are positioned not more than 1.7 m above the finished floor.

Note : This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

2: Window operation

Instructions for window operation are provided through signage or other communications to regular occupants to indicate the following:

- a. Windows with low openings are to be used during mild and/or warm weather.
- b. Windows are not to be opened when mechanical cooling is in operation (not required if no mechanical cooling is present or if mechanical cooling system is configured to disengage automatically when windows are open).
- c. Windows with high openings (if present) are to be used in cold weather.

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T09 B OUTDOOR THERMAL COMFORT | O (MAX : 3 PT)

Intent : Address the thermal comfort needs of project occupants in the exterior spaces of the project.

Summary : This WELL feature requires projects to design outdoor spaces to avoid excessive wind and manage elevated temperatures through shading or other strategies.

Issue : As communities expand, development takes the place of natural features of the landscape. By replacing open spaces and vegetation with structures built of materials that retain heat, a phenomenon known as the urban heat island has developed in the urban landscape.^{1,2} The most prominent feature of an urban heat island is a sustained elevated temperature when compared to the immediately surrounding suburban or undeveloped areas.³ This can lead to adverse health outcomes such as heat-related illness, fatigue, increased breathing and increased heart rate.^{1,4} In addition, high-rise buildings also can concentrate prevailing winds to higher speeds that result in pedestrian discomfort and dangerous conditions.⁵

Solutions : Various strategies can be used to mitigate urban heat island effects and improve outdoor thermal comfort, including adding vegetation, providing shading structures, increasing reflectance of building materials and introducing water features.⁶ The use of trees or vegetation facilitates evapotranspiration and provides shade, contributing to lower surface and air temperatures.¹ Co-benefits of introducing urban vegetation and green spaces include improved air quality, water quality (often via improved stormwater management) and savings in energy.⁷ Shelter canopies address solar radiation and provide relief to thermal stress in extreme temperatures. A shaded surface can be up to 25 °C [45 °F] cooler than the peak temperature of a nearby unshaded surface.⁷ Reducing surface temperatures of sidewalks, roadways and parking lots can be effectively achieved through a mixture of shading and high-albedo paving materials.^{8–10} Finally, water bodies can have a positive effect on microclimates in urban areas by moderating extreme temperatures. Small water bodies such as fountains, pools and ponds can regulate temperature fluctuations and help improve thermal comfort during days of extreme heat.^{11,12} Studies suggest that temperatures around and downwind of water bodies can be 1–2°C [2–4°F] lower compared to surrounding areas.¹² Computational fluid dynamics programs and other modeling tools can be used to predict the thermal conditions and likelihood of excessive wind and the effectiveness of countermeasures.^{5,13}

PART 1 MANAGE OUTDOOR HEAT (MAX : 1 PT)

For All Spaces:

Option 1: Outdoor shading

The following areas (if present) are shaded for more than half of daylight hours each day by tree canopies, awnings, or other structures:

- a. At least 50% of pedestrian pathways and building entrances.
- b. At least 25% of parking spaces (if present).
- c. At least 25% of all plazas, seating areas, exercise facilities with a contiguous area of less than 230 m² and other outdoor areas of congregation.

OR

Option 2: Temperature modeling

For pedestrian pathways and building entrances, parking spaces, and plazas, seating areas, exercise facilities with a contiguous area of less than 230 m² and other outdoor areas of congregation, project provides the following:

- a. Highest expected measure of thermal perception for each month (e.g., highest Physiologically Equivalent Temperature, highest Universal Thermal Climate Index).
- b. If the highest measure of thermal perception is associated with “moderate” (or more severe) heat stress, a list of countermeasures within at least two of the following categories and the expected reduction in heat stress that they provide:
 1. Landscaping and greenery.
 2. Manufactured shading systems (e.g., canopies).
 3. Reflectance of manufactured surfaces (e.g., sidewalks, rooftops).
 4. Water features (e.g., ponds, fountains).

Note :

Interior projects may achieve this part if the grounds of the base building meet the requirements, even if they are outside of the interior project boundary.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 AVOID EXCESSIVE WIND (MAX : 1 PT)

For All Spaces:

A computational fluid dynamic model of the building and any adjacent buildings that takes into account at least one day per season (i.e., per quarter) demonstrates the following:

- a. Winds are not expected to exceed 5 m/s for more than 5% of hours in the year in seating areas or 10% of hours on paths and parking lots.¹⁴
- b. Winds are not expected to exceed 15 m/s on paths, parking lots or seating areas for more than 0.05% of hours in the year.¹⁴

Note :

Interior projects may achieve this part if the grounds of the base building meet the requirements, even if they are outside of the interior project boundary.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 3 SUPPORT OUTDOOR NATURE ACCESS (MAX : 1 PT)

For All Spaces:

Project achieves the following features:

- a. Feature T09β Outdoor Thermal Comfort, Part 1 or Part 2.
- b. Feature M09, Part 2: Provide Nature Access Outdoors.

Note : This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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APPENDIX T1:

Surveys may be administered via paper-based surveys distributed to occupants or through a digital survey such as a website or mobile application. It is recommended that surveys are administered during mid-morning or mid-afternoon, at least 30 minutes after arrival or after a lunch break.

Survey Questionnaire:

Please answer the following questions regarding your general thermal experience during the current heating/cooling season.

1. Please rate your satisfaction level with the thermal environment in this space:

- ☐ Very Dissatisfied
- ☐ Dissatisfied
- ☐ Slightly Dissatisfied
- ☐ Neutral
- ☐ Slightly Satisfied
- ☐ Satisfied
- ☐ Very Satisfied

2. Would you prefer this area to be:

- ☐ Cooler
- ☐ Slightly Cooler
- ☐ No Change Required
- ☐ Slightly Warmer
- ☐ Warmer

If your answer to Question 1 contains the word "Dissatisfied," please answer the following two questions:

3. How do you generally perceive the thermal environment in this area:

- ☐ Cold
- ☐ Cool
- ☐ Slightly Cool
- ☐ Neutral
- ☐ Slightly Warm
- ☐ Warm
- ☐ Hot

4. Please mark the potential source (or sources) of your thermal discomfort:

- ☐ Inappropriate room thermostat setpoint temperature
- ☐ Thermostat setpoint temperature being adjusted by occupants
- ☐ Daily variations in room temperature (such as higher temperature in the afternoons)
- ☐ Occasional variations in room temperature (such as being occasionally warm or cold)
- ☐ Parts of the room being too hot
- ☐ Parts of the room being too cold
- ☐ Humidity level is too high
- ☐ Dry air
- ☐ Air movement is too high

- ☐ Air movement is too low
- ☐ Hot/cold air coming from windows
- ☐ Hot/cold walls
- ☐ Solar radiation
- ☐ Local thermal discomfort (part/parts of the body being cold or hot)
- ☐ Heat from equipment and appliances
- ☐ Strict dress code
- ☐ Hot/cold furniture surfaces
- ☐ Furniture causing hot/cold thermal discomfort or sweating
- ☐ Others

Example of Result Analysis:

The Percentage of Satisfaction should be reported as the percentage of people who are satisfied with the thermal environment of the space as shown in the following equation:

$$\text{Percentage of Satisfaction} = (\text{Number of Satisfied Occupants}) / (\text{Number of Surveyed Occupants}) \cdot 100$$

Occupants voting for "Neutral," "Slightly Satisfied," "Satisfied," and "Very Satisfied" should be deemed satisfied with the room's thermal environment. Table 1 shows the results from a typical survey and Table 2 depicts a simple example of the survey report. The results of question 2 (and question 3 if applicable) should be submitted as the number of votes for each thermal sensation and preferences condition as specified in Table 2.

Table 1: Results from a Typical Thermal Comfort Satisfaction Survey with 15 Participants

	Question 1	Question 2	Question 3	Question 4
Subject 1	Very Satisfied	No Change		
Subject 2	Satisfied	Slightly Cooler		
Subject 3	Slightly Dissatisfied	Warmer	Cold	Parts of the room being too cold
Subject 4	Slightly Dissatisfied	Slightly Warmer	Cool	Hot/Cold air coming from windows
Subject 5	Satisfied	No Change		
Subject 6	Very Satisfied	No Change		
Subject 7	Neutral	Slightly Warmer		
Subject 8	Very Satisfied	No Change		
Subject 9	Dissatisfied	Warmer	Slightly Cool	Inappropriate room thermostat setpoint temperature
Subject 10	Slightly Satisfied	Slightly Cooler		
Subject 11	Neutral	Slightly Warmer	Slightly Cool	Hot/Cold air coming from windows
Subject 12	Slightly Satisfied	Slightly Warmer		
Subject 13	Very Satisfied	No Change		
Subject 14	Slightly Satisfied	Slightly Warmer		
Subject 15	Satisfied	No Change		

Table 2: A typical Thermal Comfort Satisfaction Survey Report

1. Please rate your satisfaction level with the temperature in this room:

Considered Dissatisfied			Considered Satisfied			
Very Dissatisfied	Dissatisfied	Slightly Dissatisfied	Neutral	Slightly Satisfied	Satisfied	Very Satisfied
0	1	2	2	3	3	4

Percentage of Occupant Satisfaction:

80%

2. Would you prefer this area to be (based on the number of votes):

Cooler	Slightly Cooler	No Change	Slightly Warmer	Warmer
0	2	6	5	2

3. How do you generally perceive the thermal environment in this area (based on the number of votes):

Cold	Cool	Slightly Cool	Neutral	Slightly warm	Warm	Hot
1	1	2	0	0	0	0